Sesi 9

28 April 2026

Aggregate Demand and Aggregate Supply

Pengantar Ekonomi 2

Sumber: Mankiw (Ch 34)

Nanda Alfarina

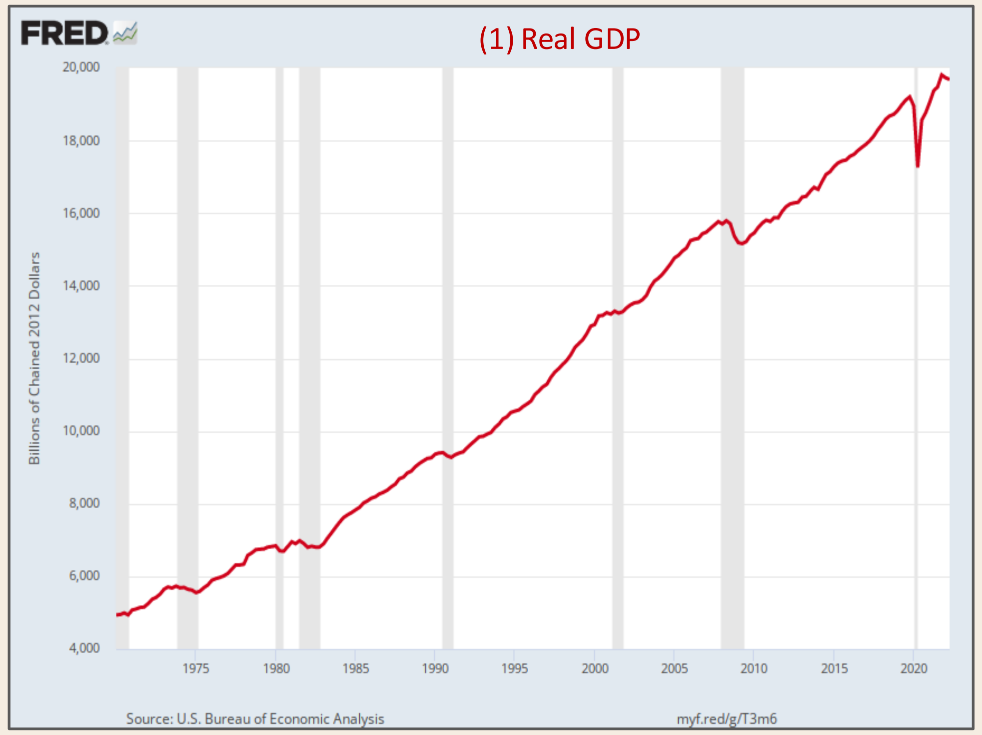
What is Economic Fluctuations?

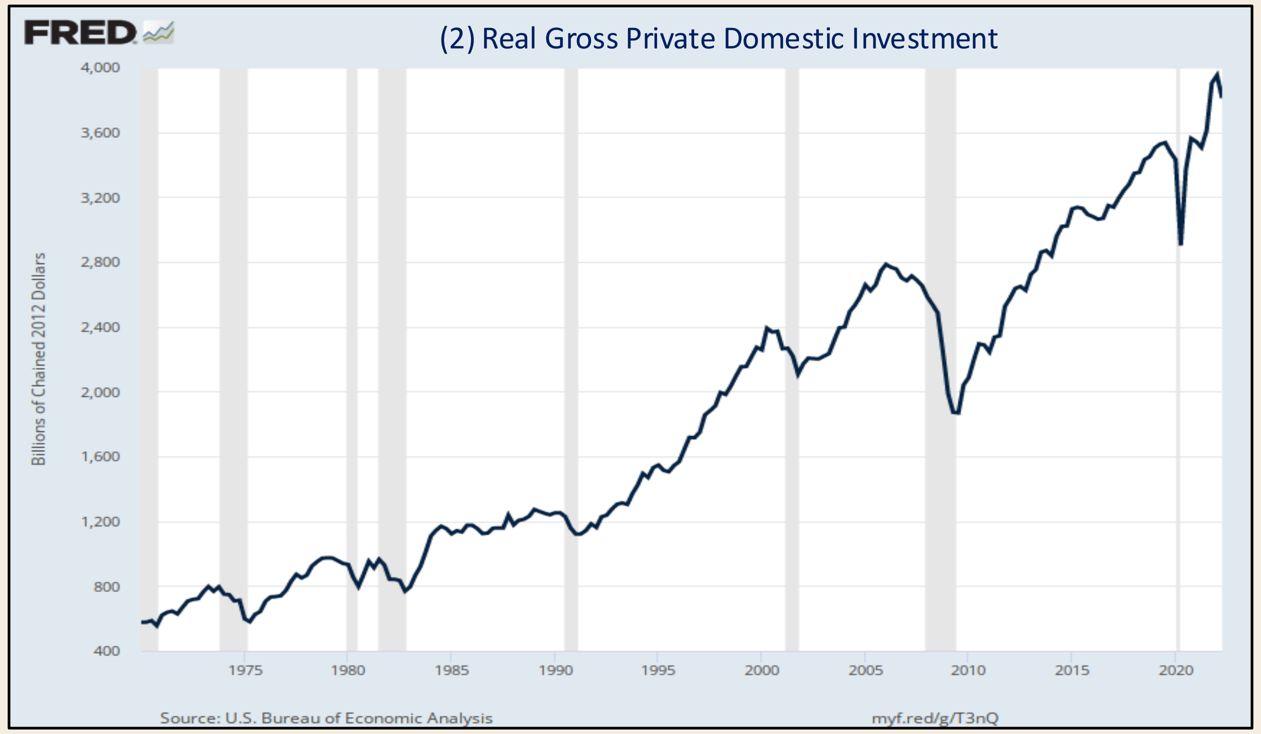
Fluctuations

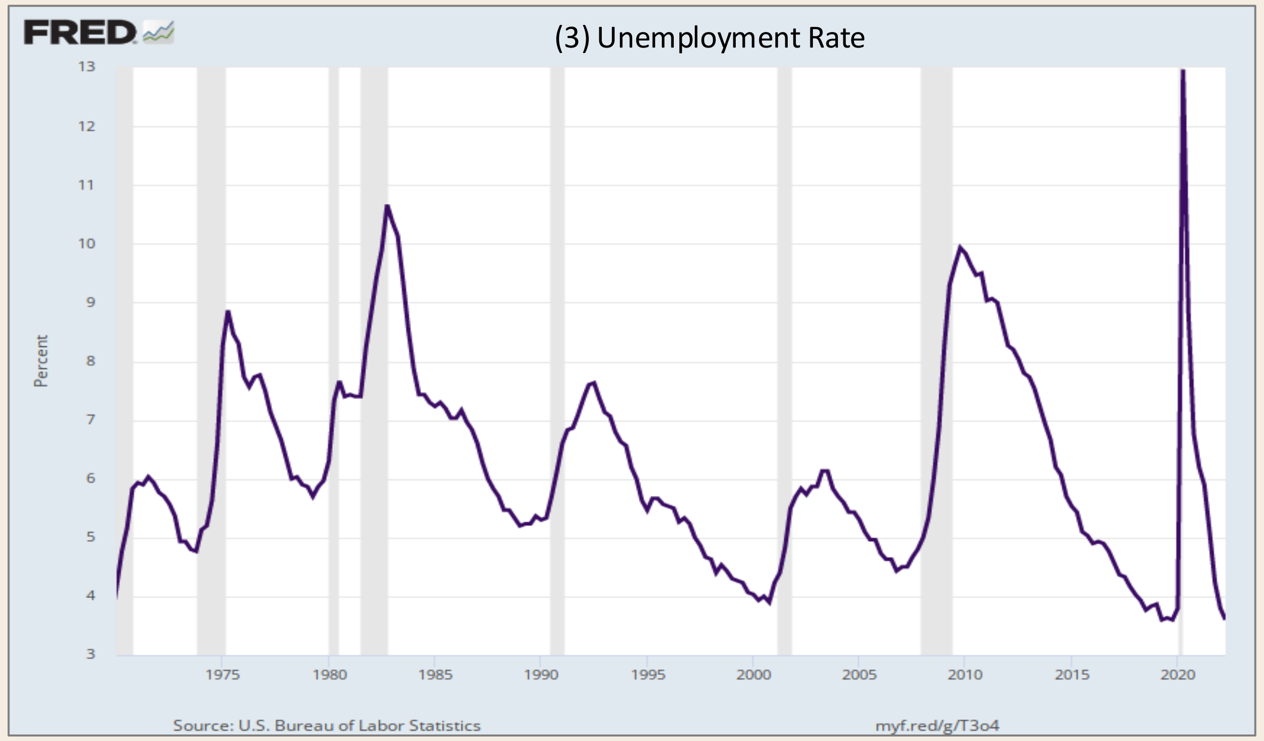
- Recession: Periods of **mild** falling incomes and rising unemployment.
- Depression: Periods of **severe** falling incomes and rising unemployment.

3 key facts about economic fluctuations

- Irregular and Unpredictable
- Most macroeconomic quantities together
- As output falls, unemployment rises







Explanations of short-run economic fluctuations

Assumptions of Classical Economics

Classical Dichotomy:

- **Real variables** → quantities/relative price
- **Nominal variables** → in terms of money

- Changes in money supply affect nominal variables but not real variables.
- If money supply doubles, everything would cost twice as much, but income would increase twice as well.

However...

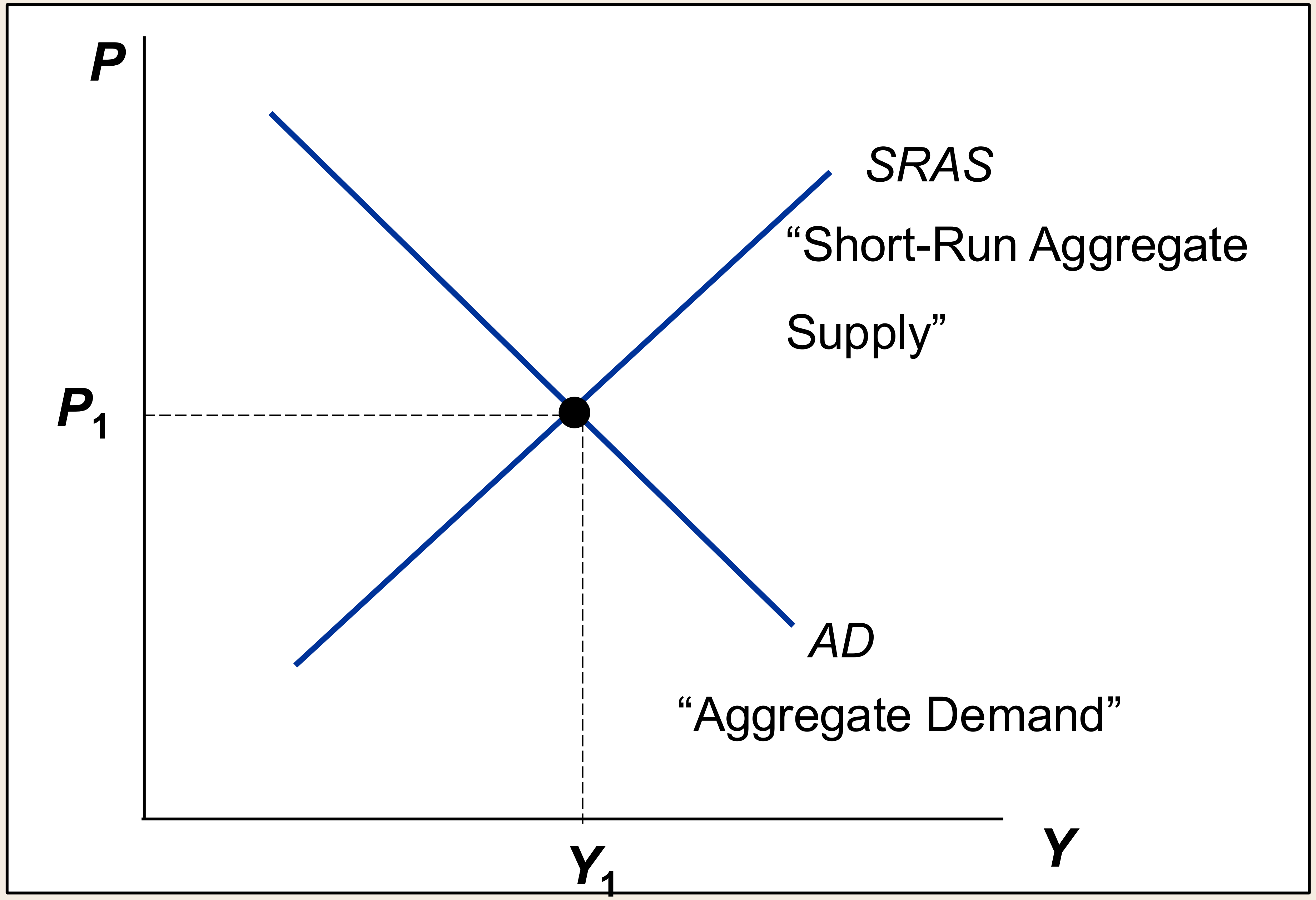
Reality

Most economists believe that the classical dichotomy describes the world in in the long run but not in the short run.

When studying year-to-year challenges, monetary neutrality is no longer appropriate as real and monetary as real and nominal variables highly intertwined.

Aggregate Demand and Aggregate Supply

To model short-run economic fluctuations, we use AD and AS.



Explanation

We use 2 variables
Y: Real GDP, output of goods and services
P: Average level of prices (e.g, CPI/GDP deflator).

Aggregate-demand → quantity of goods and services that households, firm, the government, and customers abroad wants to buy at a certain price level.
Y = C + I + G + NX

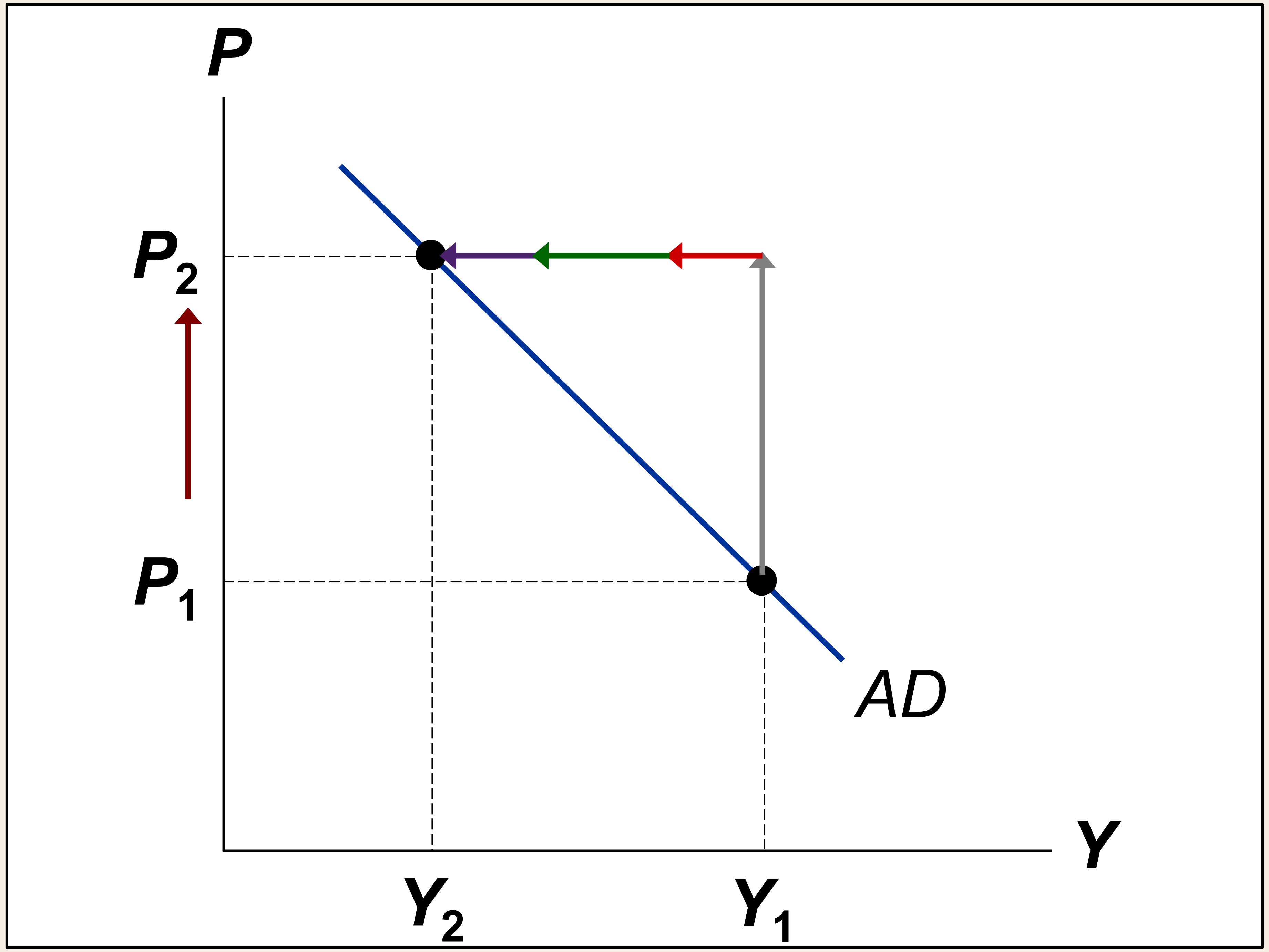
Aggregate-supply → quantity of goods and services that firms want to sell at a certain price level.

However, this model is actually different from the demand and supply curve.

Aggregate Demand

Why the AD curve slope downwards?

AD slopes downward as decreases in price level would increase the quantity of goods and services demanded.



Explanation

Wealth effect → *price level* turun akan meningkatkan nilai uang, menyebabkan seseorang bisa membeli lebih banyak barang dibanding sebelumnya, meningkatkan kemampuan untuk membeli barang dan jasa. Vice versa.

Interest-rate effect → *price level* yang lebih rendah akan menurunkan *interest rate*, meningkatkan investasi, sehingga meningkatkan permintaan barang dan jasa. Vice versa.

Exchange-rate effect → turunnya *price level* menurunkan *interest rate*, sehingga *real value* dari mata uang turun di *foreign currency exchange market*. Depresiasi yang terjadi akan meningkatkan NX dan permintaan akan barang dan jasa.

Summing Up There are three distinct but related reasons a fall in the price level increases the quantity of goods and services demanded:

1. Consumers become wealthier, stimulating the demand for consumption goods.
2. Interest rates fall, stimulating the demand for investment goods.
3. The currency depreciates, stimulating the demand for net exports.

Why the ***AD*** Curve Might Shift?

Pergeseran kurva AD disebabkan perubahan di C,I,G, dan NX pada tingkat harga tertentu.

Changes in C

- Stock market boom/crash
- Preferences re: consumption/saving tradeoff
- Tax hikes/cuts

Changes in *G*

- Federal spending, e.g., defense
- State & local spending, e.g., roads, schools

Changes in *I*

- Firms buy new computers, equipment, factories
- Expectations, optimism/pessimism
- Interest rates,
- Monetary policy,
- Tax policy or other tax incentives

Changes in *NX*

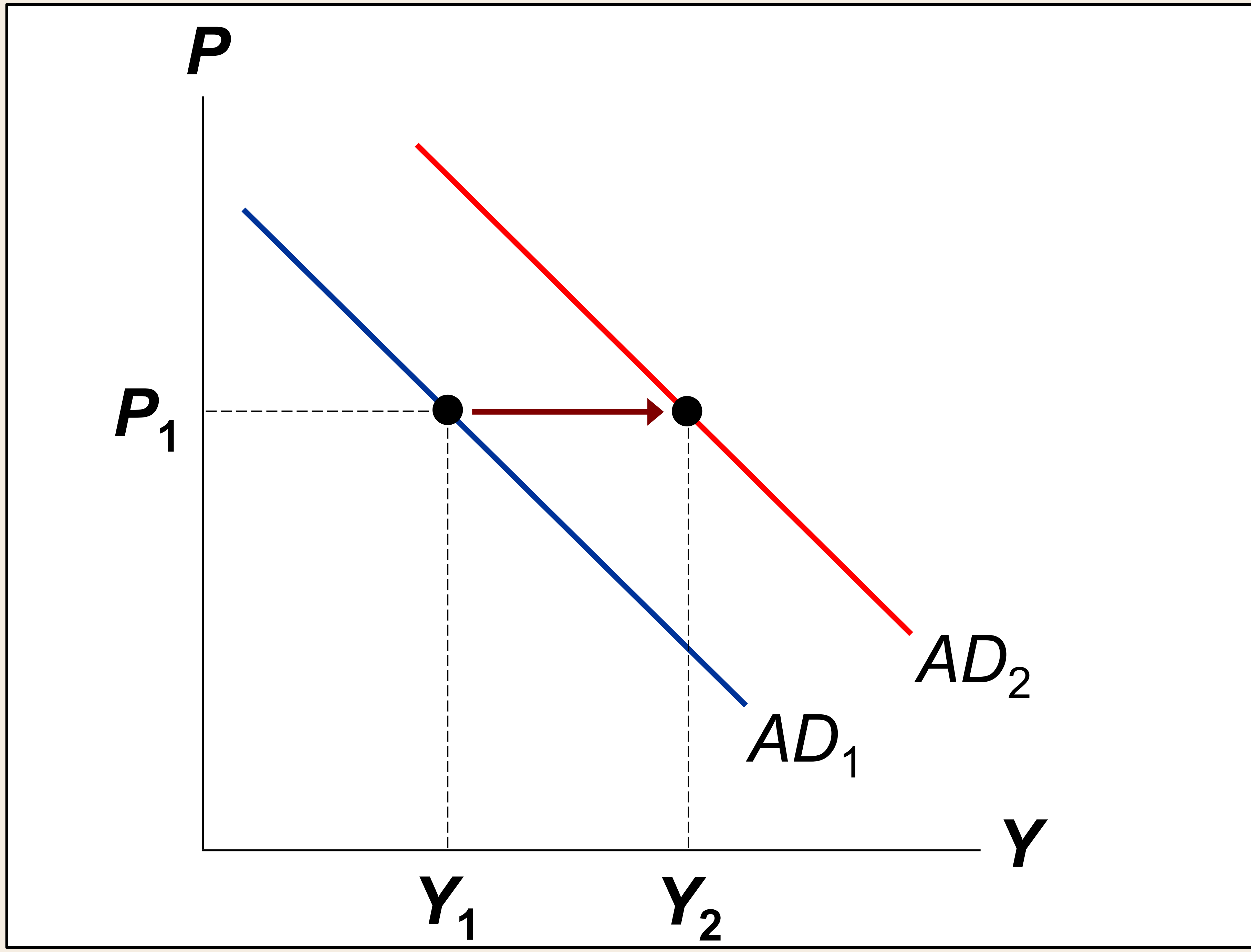
- Booms/recessions in countries that buy our exports
- Appreciation/depreciation resulting from international speculation in foreign exchange market

EXAMPLE 1: A shift in the *AD* curve

What is the effect of a stock market boom on the ***AD*** curve?

A stock market boom makes households feel wealthier, ***C*** rises, the ***AD*** curve shifts right.

Any event that changes ***C***, ***I***, ***G***, or ***NX*** (except a change in ***P***) will shift the ***AD*** curve.



Active Learning 1: The aggregate-demand curve

What happens to the ***AD*** curve in each of the following scenarios?

A. A ten-year-old investment tax credit expires.

B. The U.S. exchange rate falls.

C. A fall in prices increases the real value of consumers' wealth.

D. State governments replace their sales taxes with new taxes on interest, dividends, and capital gains.

Answers:

A. ***I*** falls, ***AD*** curve shifts left

B. ***NX*** rises, ***AD*** curve shifts right

C. Move down along ***AD*** curve (wealth-effect)

D. ***C*** rises, ***AD*** shifts right.

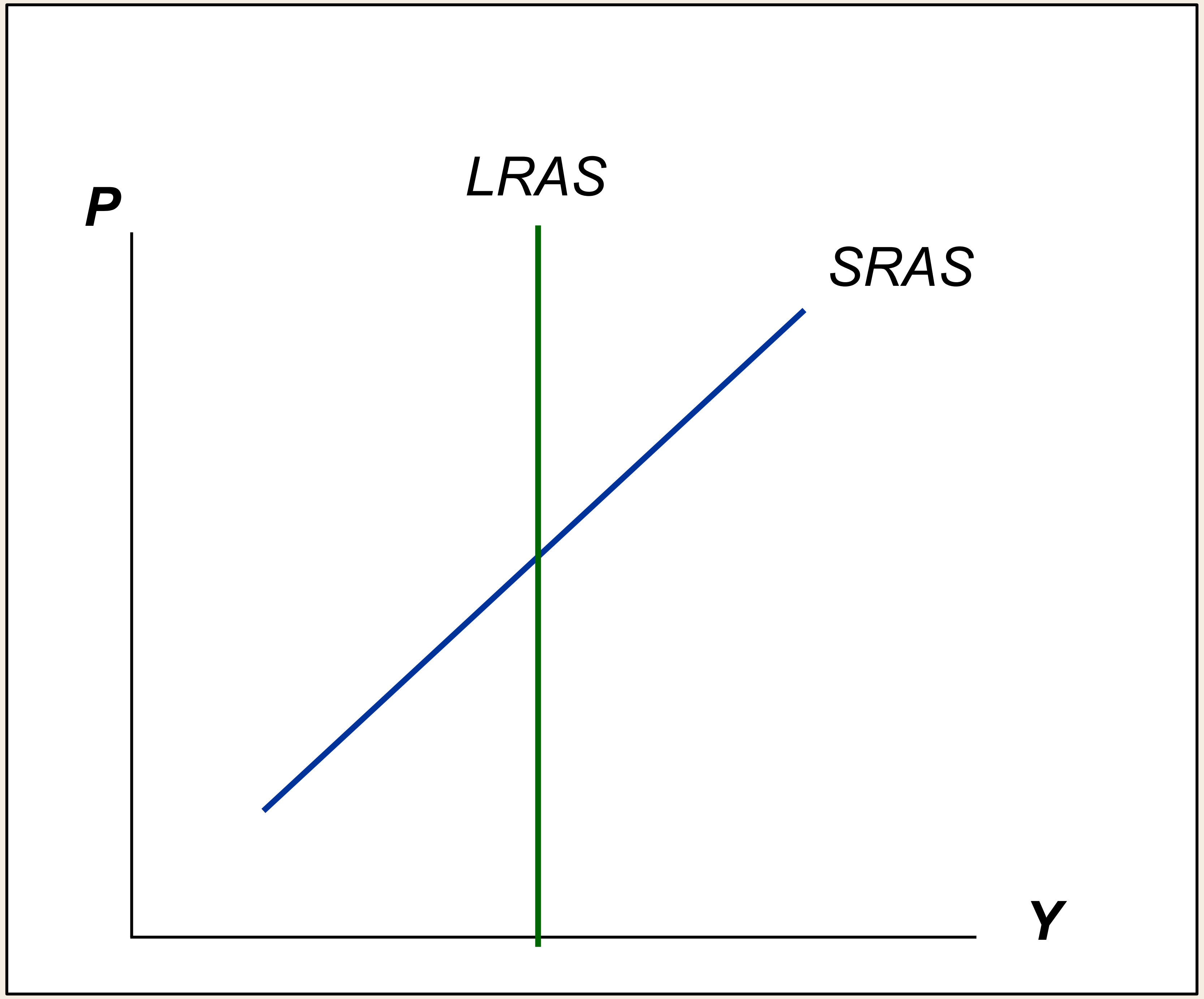
Aggregate Supply

The aggregate-supply (*AS*) curves

The ***AS* curve** shows the total quantity of goods and services firms produce and sell at each price level.

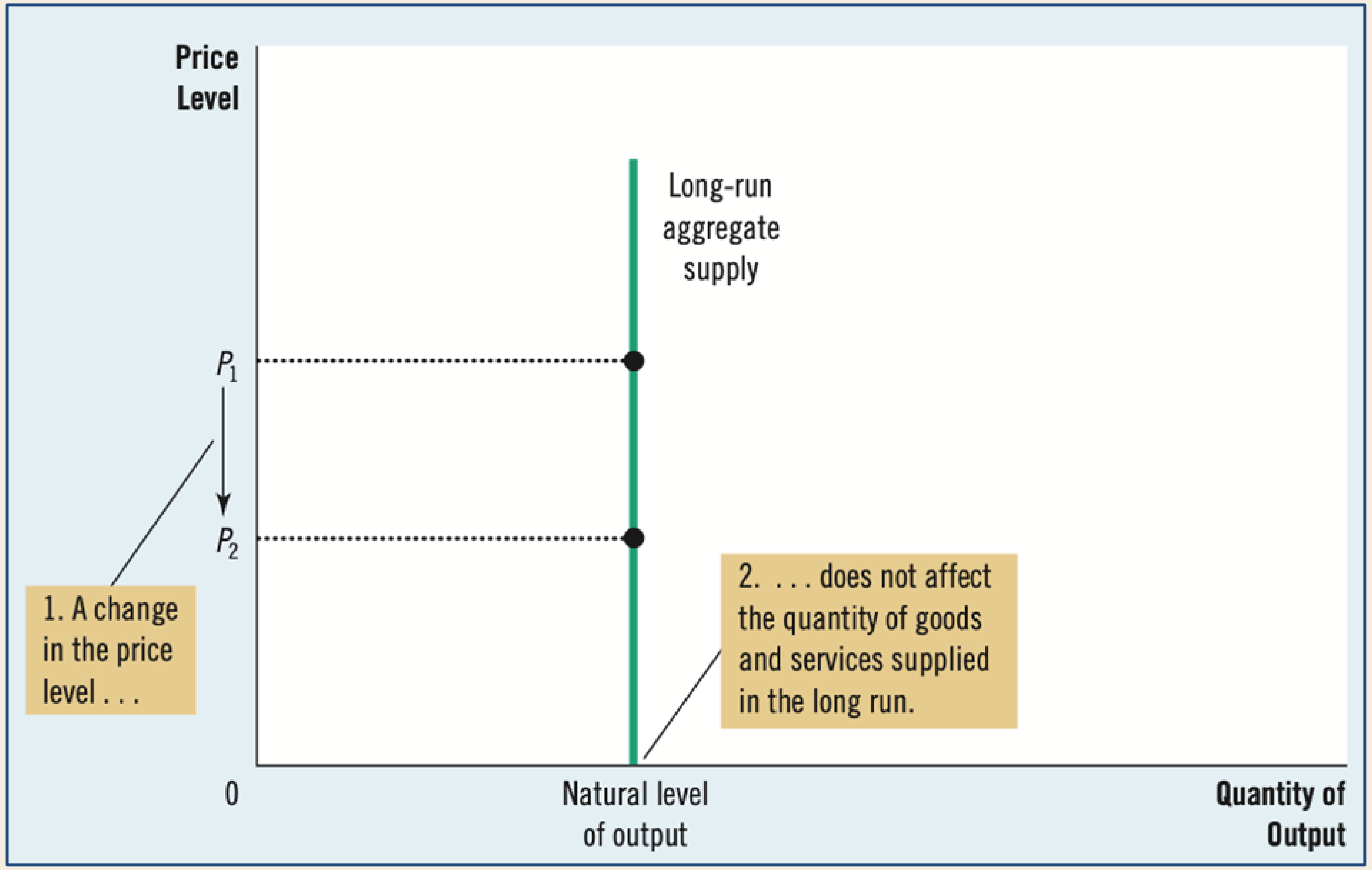
AS is:

- upward-sloping in short run
- vertical in long run



Why LRAS is vertical?

*In the long run, an economy's production of goods and services (its real GDP- **Y_N**) depends on its supplies of labor, capital, natural resources and on the available technology used to turn these factors of production into goods and services. LRAS doesn't depend on price level, so* An increase in ***P*** does not affect any of these, so it does not affect ***Y_N*** *(Classical dichotomy)*



Shifts in LRAS

Shifts arising from changes in labor → increases in number of labor would increase quantity of goods and services supplied

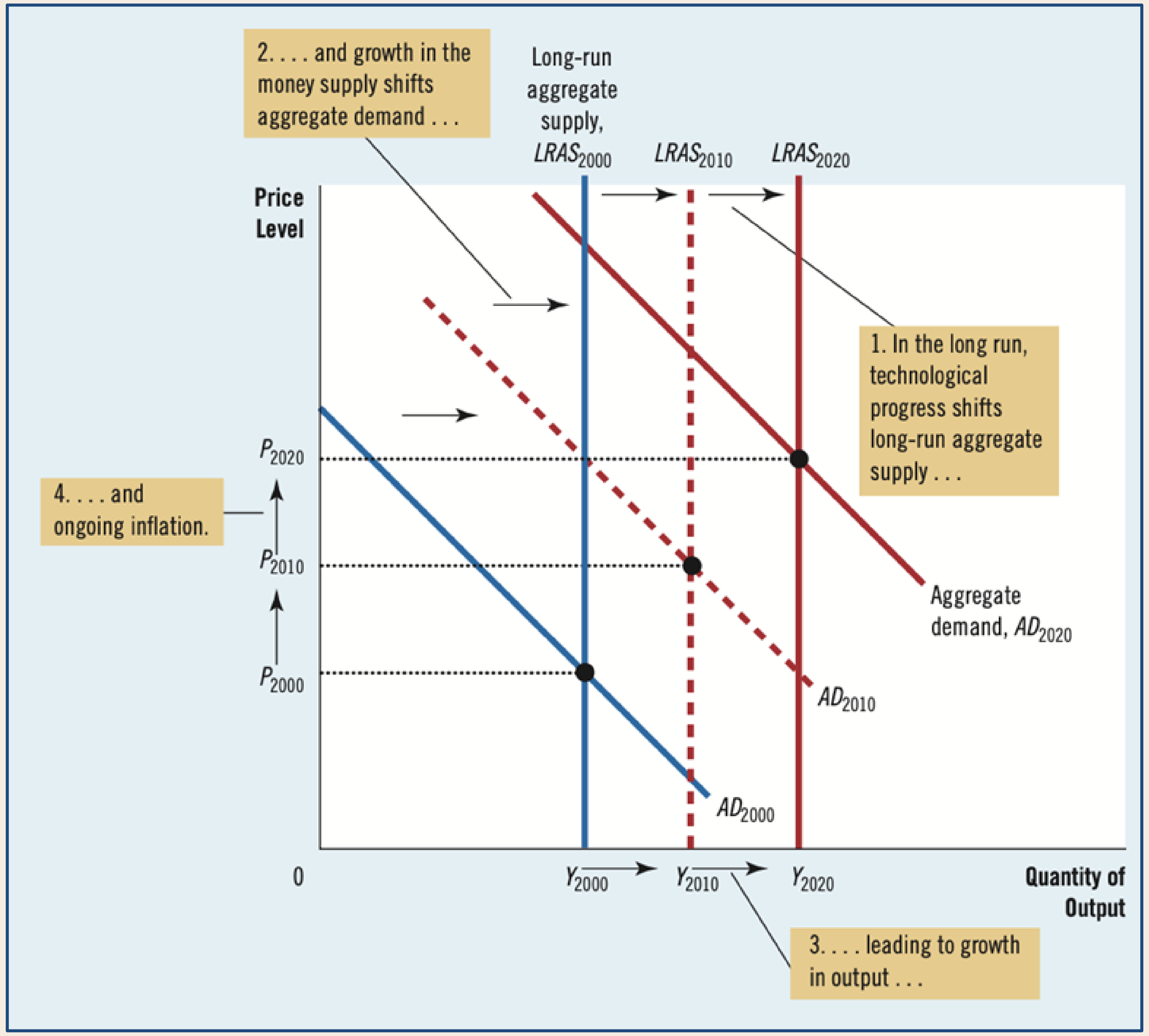
Shifts arising from changes in capital.→ Increases in capital stock increases productivity and thereby increasing quantity supplied.

Shifts arising from changes in natural resources. → changes in availability of natural resources will affect AS.

Shifts arising from changes in technological knowledge.

The **natural rate of output** (Y_N) is the amount of output the economy produces when unemployment is at its natural rate. Also called **potential output** or **full-employment output**.

Long Run Growth and Inflation



Explanation

1) Perkembangan teknologi menyebabkan kurva LRAS bergerak ke kanan.

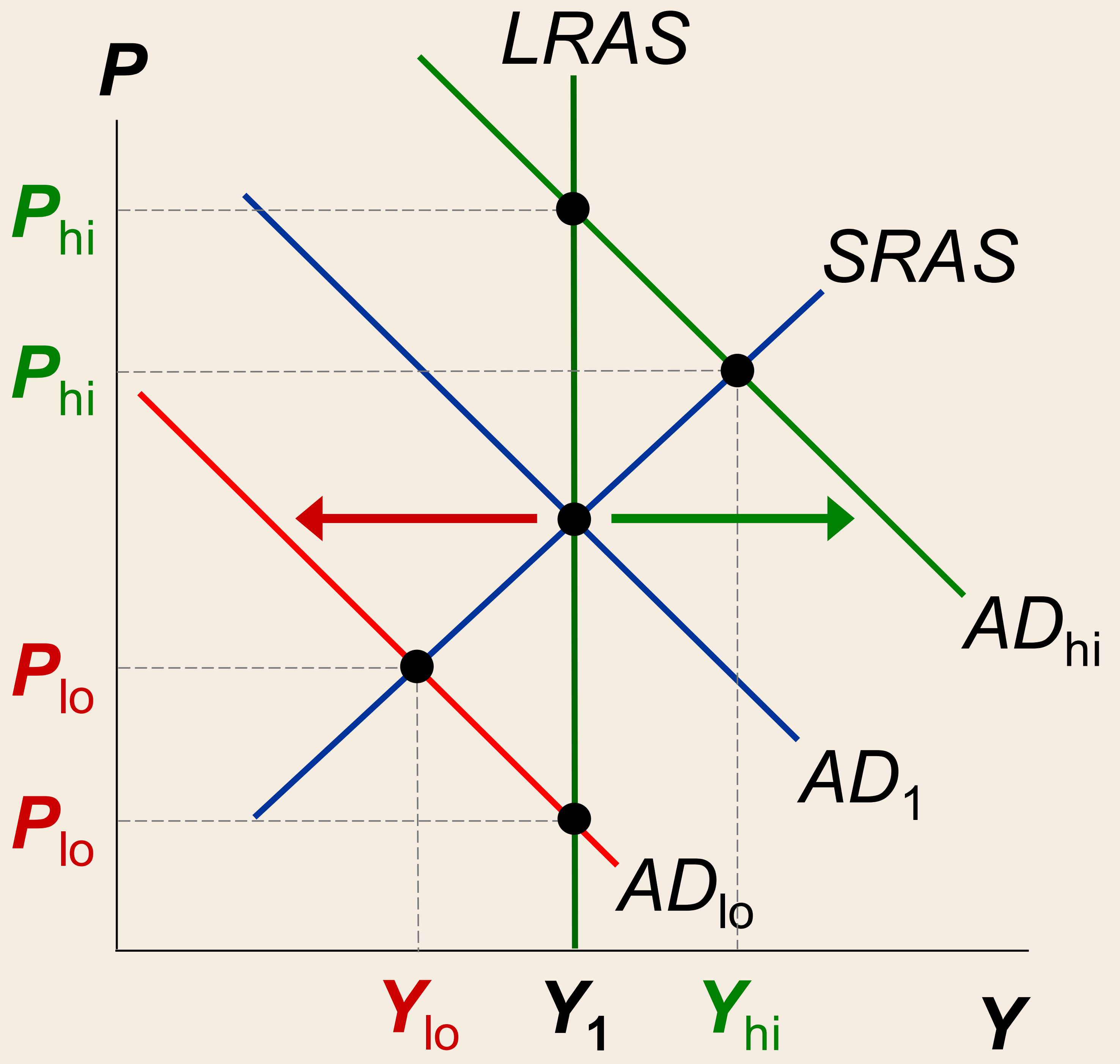
2) Pertumbuhan dalam *money supply* juga akan meningkatkan AD ke kanan.

3) Menyebabkan adanya peningkatan output dan tingkat harga meningkat.

Why the Slope of ***SRAS*** Matters

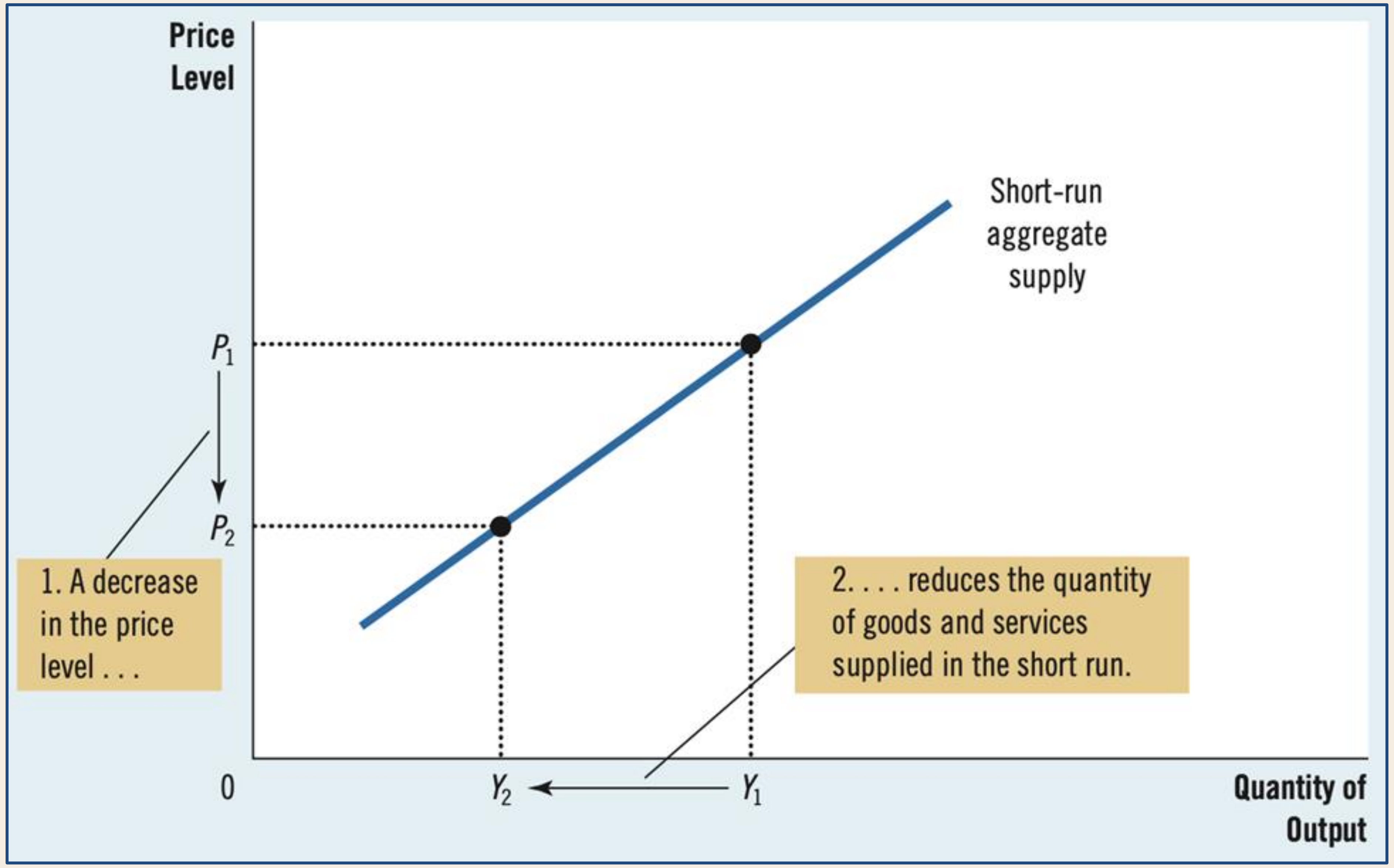
If ***AS*** is vertical, fluctuations in ***AD*** do not cause fluctuations in output or employment.

If ***AS*** slopes up, then shifts in ***AD*** do affect output and employment.



Why SRAS is upward sloping?

The short-run fluctuations in output and the price level that we will be studying should be viewed as deviations from the long-run trends of output growth and inflation. In the short run, price level affects the economy's output. The quantity of output supplied deviates from its long-run, or natural, level when the actual price level in the economy deviates from the price level that people expected to prevail.



Explanations

Sticky-wage theory → nominal wages are slow to adjust to changing economic conditions due to long-term contracts between workers and firms.
If ***P*** > ***P***$_E$, Revenue is higher, but labor cost is not. Production is more profitable, so firms increase output and employment.
*Hence, higher **P** causes higher **Y**, so the **SRAS** curve slopes upward*.

Sticky-price theory → firms are slow to readjust prices, causing a temporary increase/decrease in supply.

The misconceptions theory → changes in the overall price level can temporarily mislead economic actors.

- If ***P*** rises above ***P***$_E$
- A firm sees its price rise before realizing all prices are rising.
- The firm may believe its relative price is rising, and may increase output and employment.
- *So, an increase in **P** can cause an increase in **Y**, making the **SRAS** curve upward-sloping.*

What the 3 theories have in common

- In all 3 theories, $\boldsymbol{Y}$ deviates from $\boldsymbol{Y}_N$ when $\boldsymbol{P}$ deviates from $\boldsymbol{P}_E$.

$$Y = Y_N + a(P - P_E)$$

Quantity of output supplied

Natural rate of output (long-run)

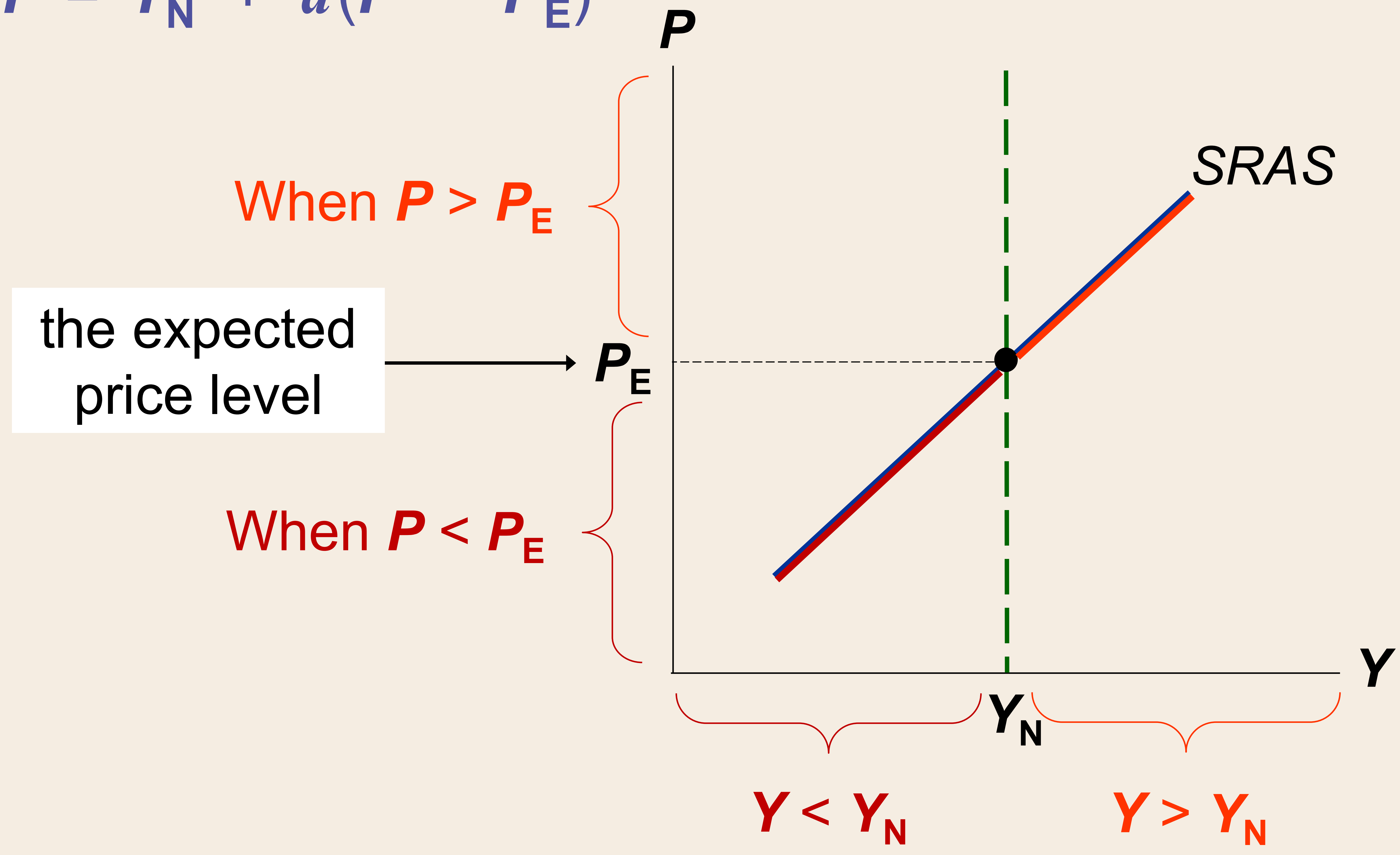
$\boldsymbol{a} > 0$, measures how much $\boldsymbol{Y}$ responds to unexpected changes in $\boldsymbol{P}$

Actual price level

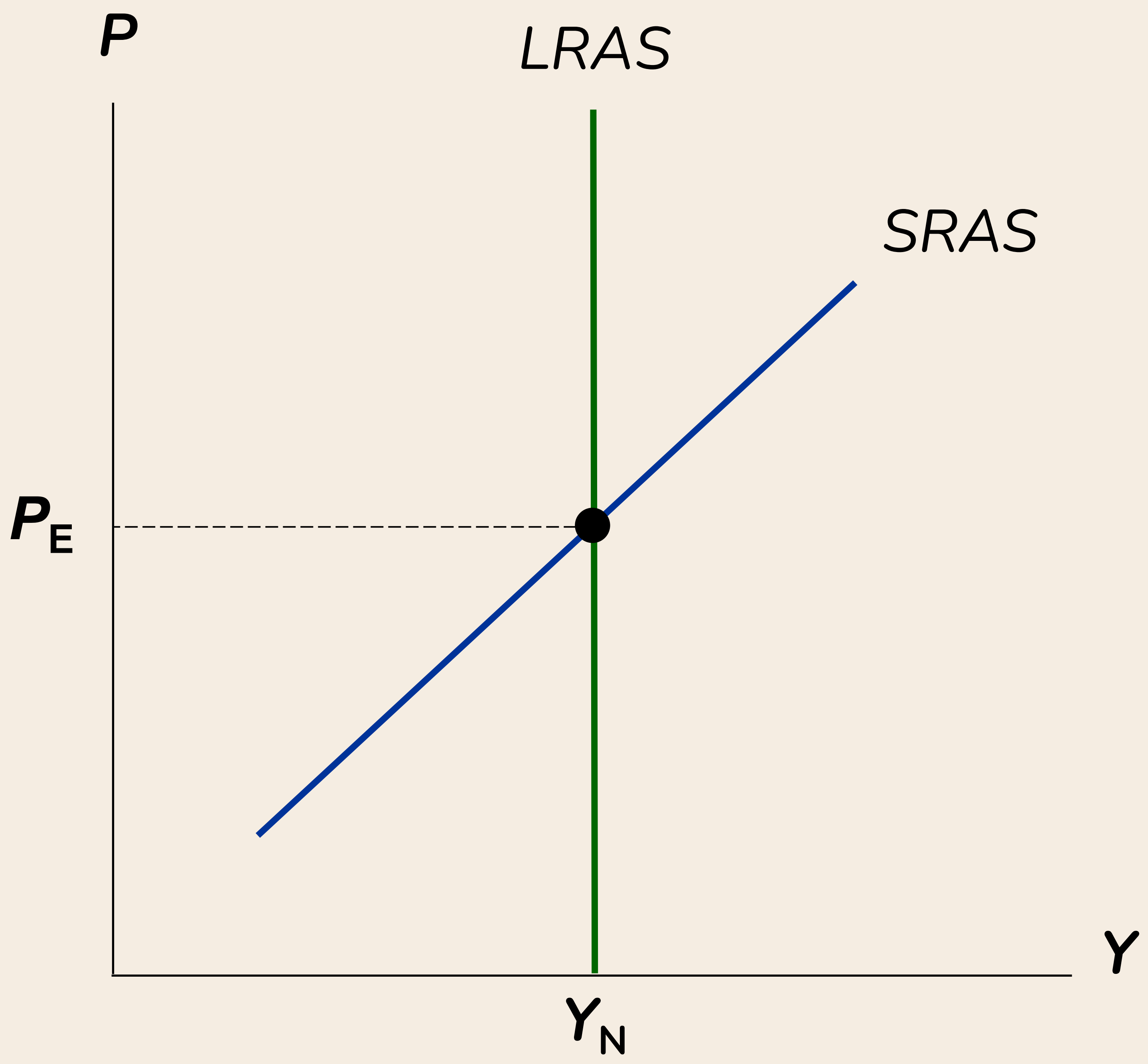
Expected price level

What the 3 theories have in common – 2

$$Y = Y_N + a(P - P_E)$$



SRAS and *LRAS*



- The imperfections in these theories are temporary. Over time,
  - Sticky wages and prices become flexible
  - Misperceptions are corrected
- In the ***LR***,
  - $P_E = P$
  - ***AS*** curve is vertical

$Y = Y_N + a(P - P_E)$

In the long run,

$P_E = P$

and

$Y = Y_N$.

and unemployment is at its natural rate.

Economic Fluctuation

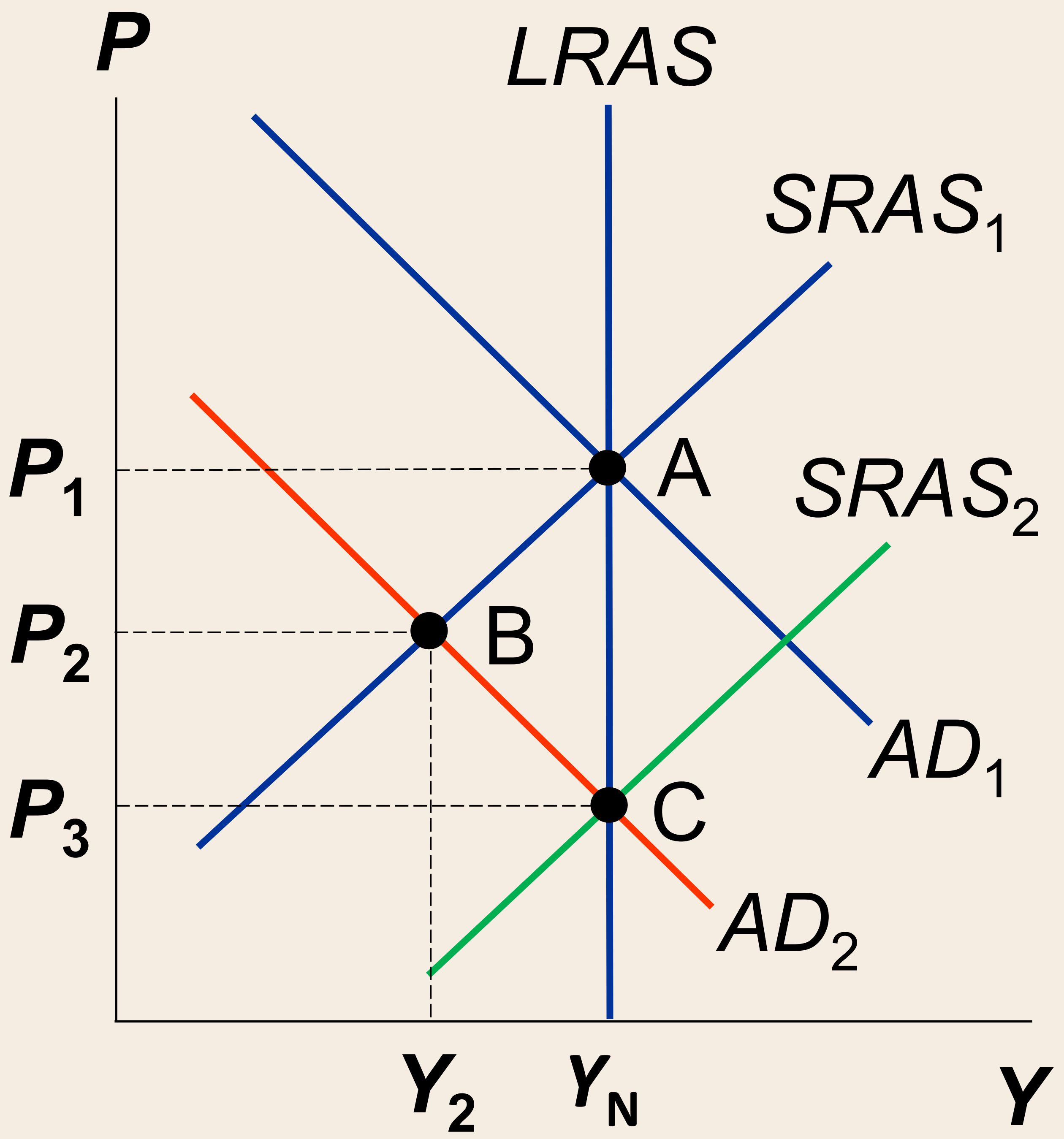
Analyzing Economic Fluctuations

- Four steps to analyzing economic fluctuations:
  1. Determine whether the event shifts ***AD*** or ***AS***.
  2. Determine whether curve shifts left or right.
  3. Use ***AD–AS*** diagram to see how the shift changes ***Y*** and ***P*** in the short run.
  4. Use ***AD–AS*** diagram to see how economy moves from new ***SR*** equilibrium to new ***LR*** equilibrium .

EXAMPLE 3: The effects of a shift in *AD*

Use the ***AD–AS*** diagram to show the effect of a stock market crash.

1. Affects ***C***, ***AD*** curve
2. ***C*** falls, so ***AD*** shifts left
3. SR equilibrium at B.
   P and ***Y*** lower,
   unemployment higher
4. Over time, P_E falls,
   SRAS shifts right,
   until LR equilibrium at C ***Y*** and
   unemployment back at initial levels.



Active Learning 2: Working with the model

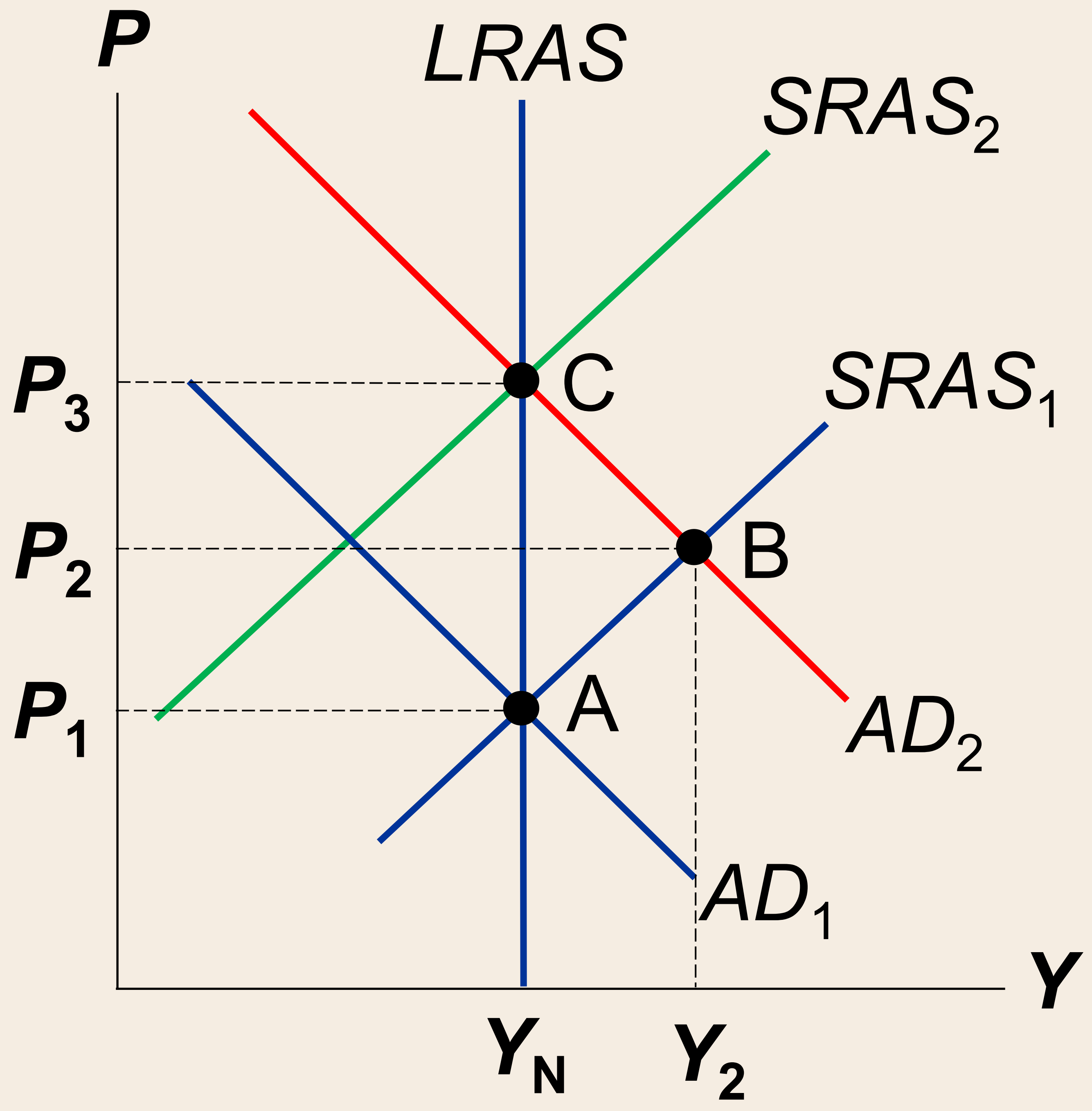
Draw the ***AD–AS*** diagram for the U.S. economy starting in a long-run equilibrium (point A).

- A boom occurs in Canada. Use your diagram to determine the SR and LR effects on U.S. ***GDP***, the price level, and unemployment.

Active Learning 2: Answers

<u>Event: Boom in Canada</u>

1. Affects ***NX***, ***AD*** curve
2. Shifts ***AD*** right
3. SR equilibrium at point B. ***P*** and ***Y*** higher, unemployment lower
4. Over time, P_E rises, ***SRAS*** shifts left, until LR equilibrium at C. ***Y*** and unemployment back at initial levels.



EXAMPLE 4: The effects of a shift in ***SRAS***

Use the ***AD–AS*** diagram to show the effect of an increase in oil prices (assume the ***LRAS*** is constant)

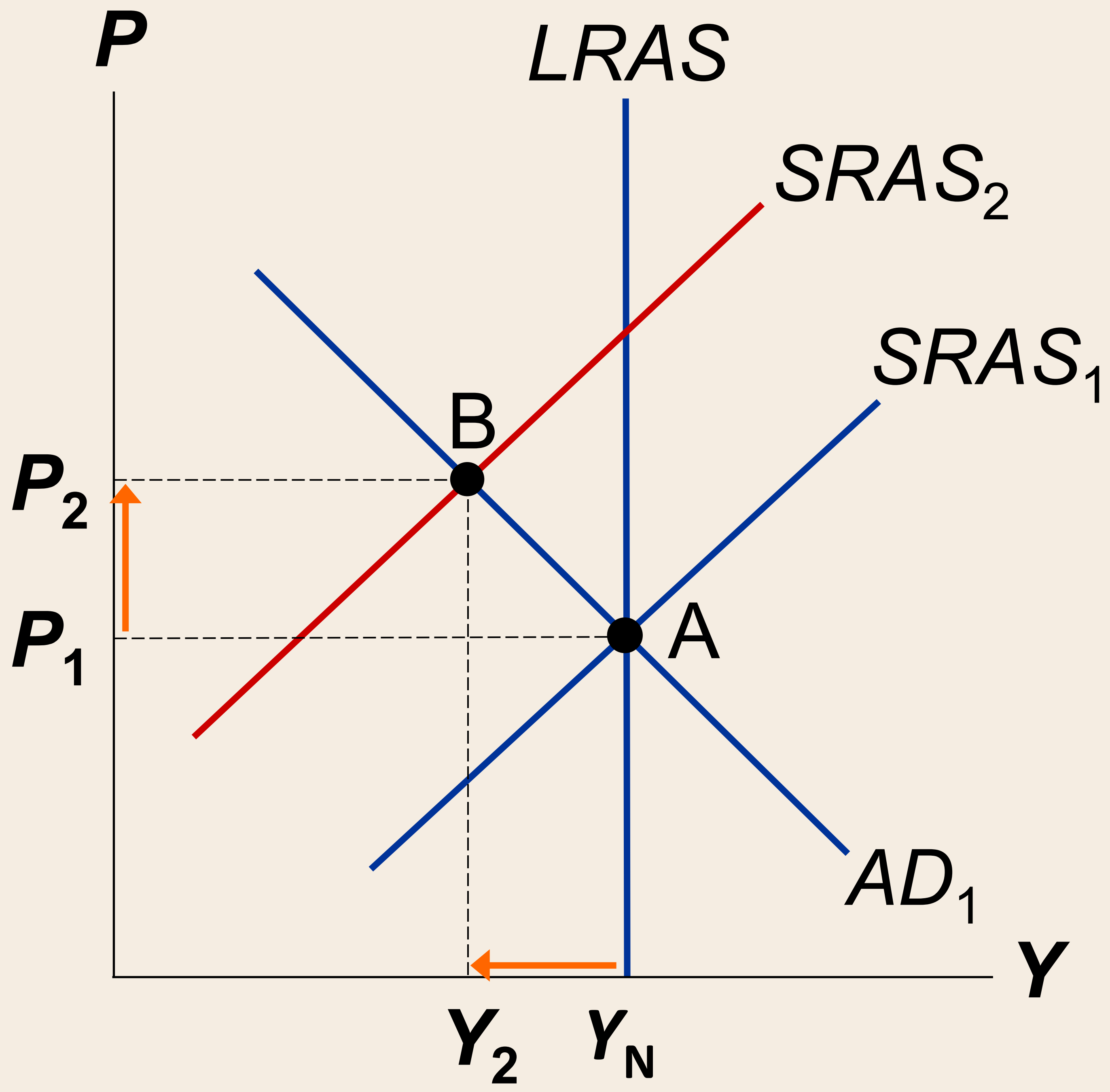
1. Increases costs, shifts **SRAS**

2. **SRAS** shifts left

3. SR equilibrium at point B.

   P higher, ***Y*** lower, unemployment higher

From A to B, **stagflation**, a period of falling output and rising prices.



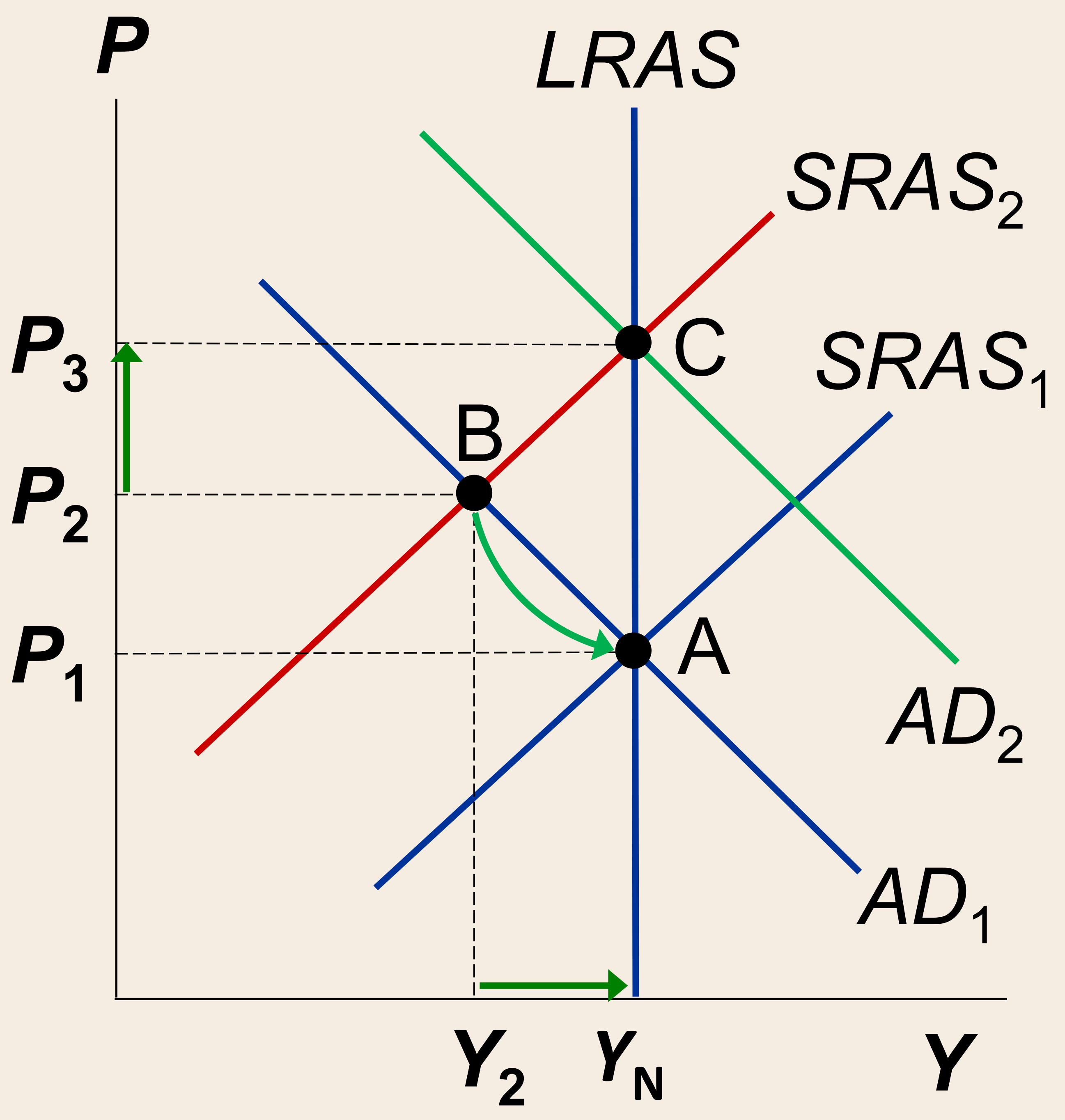
EXAMPLE 4: What happens in the long run?

If policymakers do nothing,

4. Low employment causes wages to fall, ***SRAS*** shifts right, until LR equilibrium at A.

Or, policymakers could use fiscal or monetary policy to increase ***AD*** and accommodate the ***AS*** shift:

- Y back to Y_N, but
  P permanently higher.



THINK-PAIR-SHARE

You are watching the evening news on television. The news anchor reports that union wage demands are much higher this year because the workers anticipate an increase in the rate of inflation. Your roommate says, “Inflation is a self-fulfilling prophecy. If workers think there are going to be higher prices, they demand higher wages. This increases the cost of production and firms raise their prices. Expecting higher prices simply causes higher prices.”

A. Is this true in the short run? Explain.

B. If policymakers do nothing and allow the economy to adjust to the natural level of output on its own, does expecting higher prices cause higher prices in the long run? Explain.

C. If policymakers accommodate the adverse supply shock, does the expectation of higher prices cause higher prices in the long run? Explain

Latihan Soal

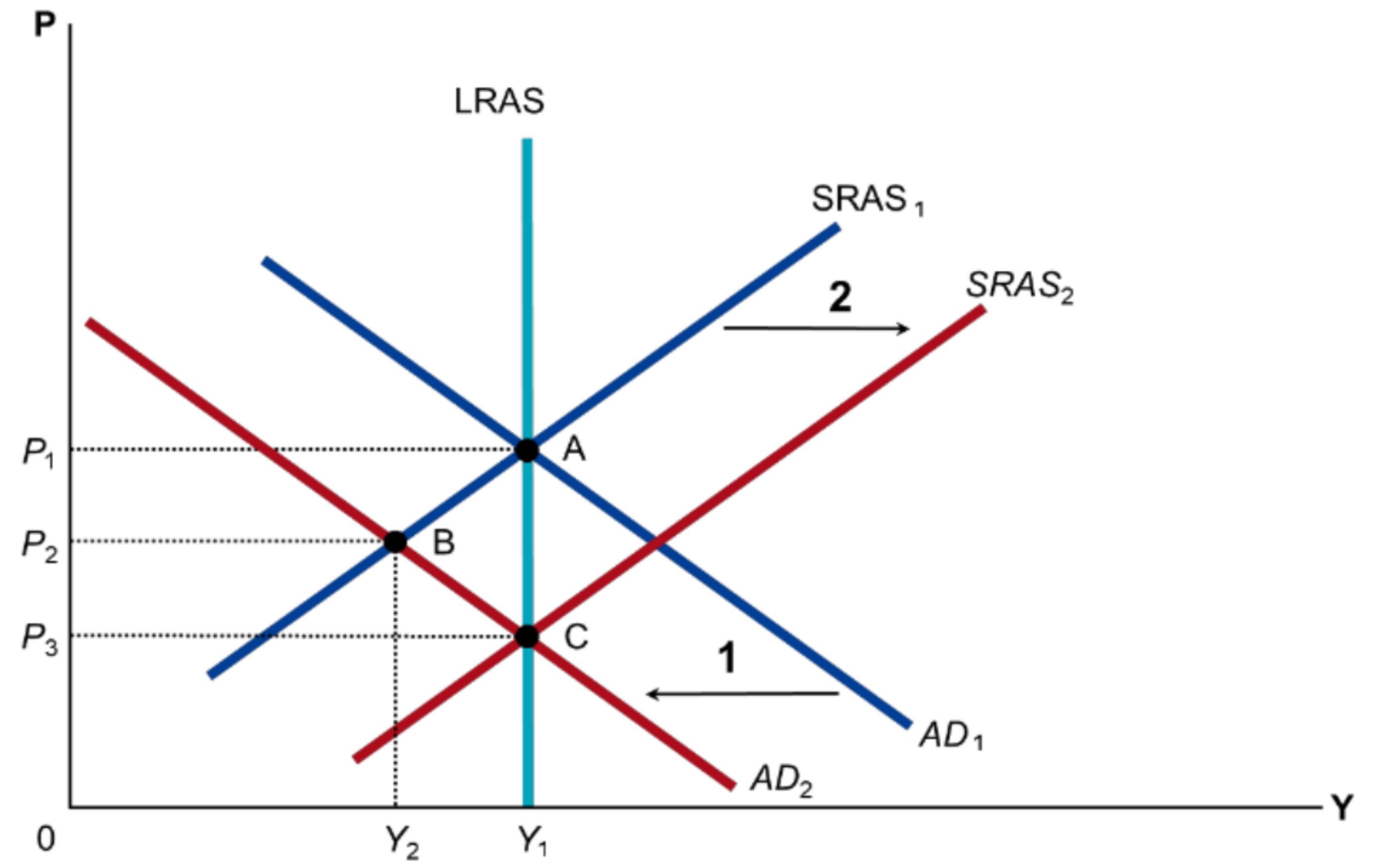
1. Hubungan negatif antara tingkat harga dan output pada kurva permintaan agregat salah satunya dijelaskan melalui *interest rate effect*, yaitu ….
   a. penurunan tingkat harga akan meningkatkan tingkat bunga, sehingga kredit investasi meningkat dan mendorong kenaikan PDB
   b. penurunan tingkat harga akan menurunkan tingkat bunga, sehingga kredit investasi meningkat dan mendorong kenaikan PDB
   c. penurunan tingkat harga akan menaikkan tingkat bunga, menyebabkan depresiasi mata uang domestik, meningkatkan ekspor neto, dan akhirnya meningkatkan PDB
   d. penurunan tingkat harga akan menurunkan tingkat bunga, menyebabkan depresiasi mata uang domestik, meningkatkan ekspor neto, dan akhirnya meningkatkan PDB

2. Penerapan tarif barang impor yang tinggi oleh Amerika terhadap barang ekspor dari Indonesia menyebabkan ....
   a. ekspor Indonesia ke Amerika menurun, mengurangi permintaan agregat, sehingga tingkat output dan harga di Indonesia cenderung turun
   b. permintaan impor Indonesia dari Amerika meningkat, meningkatkan permintaan agregat, sehingga tingkat output dan harga di Indonesia cenderung naik
   c. daya saing produk Indonesia di pasar Amerika meningkat, sehingga ekspor dan tingkat output Indonesia naik, sedangkan tingkat harganya cenderung turun
   d. ekspor Indonesia ke Amerika meningkat drastis, mendorong peningkatan permintaan agregat dan menurunkan tingkat harga di Indonesia

3. Salah satu penjelasan terkait kurva penawaran agregat dalam Makroekonomi menyebutkan bahwa kenaikan tingkat harga akan ....
   a. menaikkan upah riil jika upah nominal bersifat kaku
   b. mengurangi upah nominal jika upah riil bersifat kaku
   c. mengurangi upah riil jika upah nominal bersifat kaku
   d. menaikkan upah nominal jika upah riil bersifat kaku

4. Banyaknya penduduk yang memasuki usia pensiun (*aging population*) memberikan dampak signifikan terhadap perekonomian. Akibat dari *aging population*, maka ….
   a. kurva penawaran agregat bergeser ke kanan, sehingga tingkat harga menurun, sedangkan tingkat output dalam perekonomian meningkat
   b. kurva penawaran agregat tetap, namun permintaan agregat turun sehingga harga dan output dalam perekonomian menurun
   c. kurva penawaran agregat bergeser ke kiri, sehingga tingkat harga menurun, sedangkan tingkat output dalam perekonomian meningkat
   d. kurva penawaran agregat bergeser ke kiri, sehingga tingkat harga meningkat, sedangkan tingkat output dalam perekonomian menurun

10. Seperti terlihat pada diagram di bawah ini, pergeseran kurva permintaan agregat ke kiri dari AD_1 ke AD_2 tidak akan memengaruhi output perekonomian dalam jangka panjang karena kurva penawaran agregat kemudian akan bergeser ke kanan dari $SRAS_1$ ke $SRAS_2$. Keseimbangan perekonomian bergerak dari titik A ke titik C. Hal ini terjadi karena ....



a. penurunan permintaan agregat mendorong pemerintah meningkatkan pengeluaran, yang menyebabkan peningkatan kapasitas produksi
b. penurunan permintaan agregat menyebabkan penurunan harga, yang pada akhirnya menurunkan ekspektasi harga, sehingga penawaran agregat meningkat
c. penurunan permintaan agregat menyebabkan penurunan harga, yang mendorong peningkatan konsumsi dan investasi, sehingga penawaran agregat meningkat
d. penurunan permintaan agregat menyebabkan penurunan harga, sehingga produksi perusahaan dan konsumsi masyarakat meningkat

6. Tekanan geopolitik telah mendorong kenaikan harga energi. Kenaikan harga energi ini menyebabkan kenaikan harga barang dan jasa secara keseluruhan. Dengan menggunakan teori *Sticky-Price* pada kurva penawaran agregat jangka pendek (SRAS), jika tingkat harga naik sebesar 5% sementara ekspektasi kenaikan harga perusahaan di awal tahun hanya sebesar 3%, maka beberapa perusahaan dengan biaya menu tinggi akan memiliki ....
    a. harga relatif lebih tinggi, yang menyebabkan peningkatan jumlah agregat barang dan jasa yang ditawarkan
    b. harga relatif lebih tinggi, yang menyebabkan penurunan jumlah agregat barang dan jasa yang ditawarkan
    c. harga relatif lebih rendah, yang menyebabkan peningkatan jumlah agregat barang dan jasa yang ditawarkan
    d. harga relatif lebih rendah, yang menyebabkan penurunan jumlah agregat barang dan jasa yang ditawarkan

Esai:

1. Negara Solvaria menghadapi resesi pada tahun 2025. Untuk memulihkan perekonomian Solvaria, Pemerintah Solvaria berencana menurunkan pajak pendapatan. Dengan menggunakan model AD-AS, jelaskan bagaimana pengaruh kebijakan tersebut dalam jangka pendek.
2. Negara Niveria mengalami gagal panen karena cuaca ekstrim yang berkepanjangan. Gunakan model AD-AS untuk menjelaskan perubahan keseimbangan tingkat harga dan output di perekonomian Niveria dalam jangka pendek dan jangka panjang.

Anggaplah Negara Asgard menganut sistem perekonomian terbuka. Dengan mengasumsikan pemerintah Negara Asgard tidak melakukan intervensi, **jelaskan dan tunjukkan dampak dalam jangka pendek dan jangka panjang pada tingkat output dan harga**, jika terjadi hal-hal berikut ini.

a. Penurunan ekonomi global membuat permintaan atas barang ekspor Asgard menurun (yang ditunjukkan oleh pergeseran pada kurva permintaan agregat). (**hint**: gunakan diagram yang tepat!)

b. Peningkatan biaya produksi menyebabkan banyak industri pengolahan mengurangi jumlah produksinya (yang ditunjukkan oleh pergeseran pada kurva suplai agregat). (**hint**: gunakan diagram yang tepat!)

CHAPTER IN A NUTSHELL

- Short-run economic fluctuations around long-run trends are irregular and largely unpredictable.
- When recessions occur, real GDP and other measures of income, spending, and production fall, while unemployment rises.
- Classical economic theory assumption: nominal variables such as the money supply and the price level do not influence real variables such as output and employment. Accurate in the long run but not in the short run.
- Analyze short-run economic fluctuations: aggregate demand and aggregate supply model.
- AD-AS model: the output of goods and services and the overall level of prices adjust to balance aggregate demand and aggregate supply.
- Aggregate-demand curve slopes downward due to:

1. The wealth effect: lower price level raises the real value of households' money holdings, stimulating consumer spending.
2. The interest-rate effect: lower price level reduces the quantity of money households demand; households try to convert money into interest-bearing assets, interest rates fall, stimulating investment spending.

CHAPTER IN A NUTSHELL

3. And the exchange-rate effect: a lower price level reduces interest rates, the dollar depreciates in the market for foreign-currency exchange, stimulating net exports.

- Any event or policy that raises C, I, G, or NX at any price level increases aggregate demand.
- Any event or policy that reduces C, I, G, or NX at any price level decreases aggregate demand.
- The LRAS curve is vertical. The quantity of goods and services supplied depends on the economy's labor, capital, natural resources, and technology but not on the overall level of prices.
- Upward slope of the SRAS curve is explained by:

1. Sticky-wage theory: an unexpected fall in the price level for given nominal wages, induces firms to reduce employment and production.
2. Sticky-price theory: an unexpected fall in the price level leaves some firms with prices that are temporarily too high, reducing their sales and causing them to cut back production.
3. Misperceptions theory: an unexpected fall in the price level leads suppliers to mistakenly believe that their relative prices have fallen, inducing them to reduce production.

CHAPTER IN A NUTSHELL

- All three theories imply that output deviates from its natural level when the actual price level deviates from the price level that people expected.
- Events that alter the economy's ability to produce output, such as changes in labor, capital, natural resources, or technology, shift the short-run aggregate-supply curve (and may shift the long-run aggregate-supply curve as well). In addition, the position of the short-run aggregate-supply curve depends on the expected price level.
- Economic fluctuations caused by a shift in aggregate demand. If the AD curve shifts left, output and prices fall in the short run. Over time (change in expected price level), wages, prices, and perceptions adjust; the SRAS curve shifts right: economy back to its natural level of output at a new, lower price level.
- Economic fluctuations caused by a shift in aggregate supply. If the SRAS curve shifts left, falling output and rising prices (stagflation). Over time, wages, prices, and perceptions adjust, the SRAS curve shifts back to the right, returning the price level and output to their original levels